Demonstration: National MCP Servers for Physical AI Oncology Clinical Trial Systems

Reference Architecture Demonstration and Interactive Simulation

Kevin Kawchak

Chief Executive Officer, ChemicalQDevice

kevink@chemicalqdevice.com

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License MIT | MCP Servers 5 | Tools 23 | Tests 668 | Python 3.10 | 3.11 | 3.12

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Abstract

Prior oncology clinical trial infrastructure has been fragmented, site-specific, and inefficient across the critical dimensions of interoperability, auditability, privacy, and deployment. Each trial site has historically operated its own isolated systems for electronic health records, imaging archives, and audit logging, resulting in inconsistent data formats, duplicated regulatory effort, and limited cross-site collaboration.

This page presents a demonstration of the National MCP Physical AI Oncology Trials system, a proposed end-to-end architecture comprising five Model Context Protocol (MCP) servers that address these gaps through standardized, federated, and safety-

governed capabilities. The five servers (Authorization, FHIR Clinical Data, DICOM Imaging, Audit Ledger, and Provenance) expose 23 tools across five hierarchical conformance levels, validated by 668 test functions.

5 MCP Servers	**23** Tools
668 Test Functions	**381** Files
34 Integration Adapters	**8** Safety Modules
13 JSON Schemas	**6** Deployment Modes

Paper: 10.5281/zenodo.18916731 | Repository: kevinkawchak/national-mcp-pai-oncology-trials (DOI: 10.5281/zenodo.18894758)

Architecture Overview

Five domain-specific MCP servers form the protocol surface for Physical AI oncology trials

```
NATIONAL MCP FIVE-SERVER DEPLOYMENT
+-----------------------------------------------------+
|                  SITE DEPLOYMENT                    |
|                                                     |
|    +-----------+  +-----------+  +-----------+     |
|    | trialmcp  |  | trialmcp  |  | trialmcp  |     |
|    | authz     |  | fhir      |  | dicom     |     |
|    | (5 tools) |  | (4 tools) |  | (4 tools) |     |
|    | RBAC,     |  | FHIR R4,  |  | DICOMweb, |     |
|    | Tokens    |  | HIPAA     |  | RECIST    |     |
|    +-----+-----+  +-----+-----+  +-----+-----+     |
|          |              |              |           |
|          +------+-------+------+-------+           |
|                 |              |                   |
|          +------v------+ +-----v-------+           |
|          | trialmcp    | | trialmcp    |           |
|          | ledger      | | provenance  |           |
|          | (5 tools)   | | (5 tools)   |           |
|          | SHA-256     | | DAG, W3C    |           |
|          | 21 CFR 11   | | PROV        |           |
|          +-------------+ +-------------+           |
|                                                     |
|  23 tools total across 5 servers                    |
+-----------------------------------------------------+
```

Figure 1: Five-server deployment topology at each participating clinical site

Authorization Server 5 tools Level 1

`servers/trialmcp_authz/` | `server.py, policy_engine.py, token_store.py`

Deny-by-default RBAC, SHA-256 token lifecycle, policy evaluation

- `authz_evaluate` | `authz_issue_token` | `authz_validate_token`
- `authz_revoke_token` | `authz_list_policies`

FHIR Clinical Data Server 4 tools Level 2

`servers/trialmcp_fhir/` | `server.py, deid_pipeline.py, fhir_adapter.py`

FHIR R4 clinical data with HIPAA Safe Harbor de-identification

- `fhir_read` | `fhir_search` | `fhir_patient_lookup` | `fhir_study_status`

DICOM Imaging Server 4 tools Level 3

`servers/trialmcp_dicom/ | server.py, dicom_adapter.py`

DICOM imaging queries, RECIST 1.1 measurements

- `dicom_query` | `dicom_retrieve_pointer` | `dicom_study_metadata` | `dicom_recist_measurements`

Audit Ledger Server 5 tools Level 1

`servers/trialmcp_ledger/ | server.py, chain.py, storage.py`

Hash-chained 21 CFR Part 11 audit trail

- `ledger_append` | `ledger_verify` | `ledger_query`
- `ledger_replay` | `ledger_export`

Provenance Server 5 tools Level 4

`servers/trialmcp_provenance/ | server.py, dag.py, storage.py`

DAG-based data lineage, SHA-256 fingerprinting

- `provenance_record` | `provenance_query_forward` | `provenance_query_backward`
- `provenance_fingerprint` | `provenance_lineage`

Hash-Chained Audit Record (from `servers/trialmcp_ledger/chain.py`)

```
GENESIS_HASH = "0" * 64

def canonical_json(record: dict[str, Any]) -> str:
    filtered = {k: v for k, v in sorted(record.items())
```

```
                 if k != "hash"}
    return json.dumps(filtered, sort_keys=True,
                      ensure_ascii=True)

def compute_audit_hash(record, prev_hash: str) -> str:
    canonical = canonical_json(record)
    payload = prev_hash + canonical
    return hashlib.sha256(
        payload.encode("utf-8")).hexdigest()
```

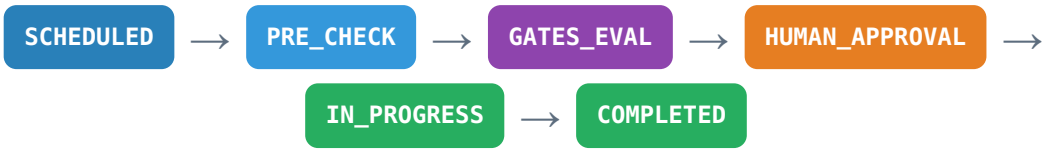
Seven Interactive Diagram Sections

Each section corresponds to a process diagram from `docs/mcp-process/`

1 Robot Procedure Lifecycle

Source: `docs/mcp-process/01-robot-procedure-lifecycle.md`

The robot procedure lifecycle defines the complete state machine for robot-assisted clinical procedures. Each state transition requires specific MCP server interactions and safety validations, with timing constraints enforcing clinical safety protocols.




```
State Transitions and MCP Server Interactions:

SCHEDULED --(pre-check triggers)--> PRE_CHECK
  Servers: trialmcp-authz (token validation)
```

```
            trialmcp-fhir  (patient eligibility)
            trialmcp-dicom (imaging availability)

PRE_CHECK --(all checks pass)--> GATES_EVAL
  Servers: safety/gate_service (5 gates evaluated)

GATES_EVAL --(all gates pass)--> HUMAN_APPROVAL
  Servers: safety/approval_checkpoint (clinician review)
  Timeout: 300 seconds

HUMAN_APPROVAL --(approved)--> IN_PROGRESS
  Servers: trialmcp-ledger (audit record)
           safety/procedure_state (state transition)

IN_PROGRESS --(procedure complete)--> COMPLETED
  Servers: trialmcp-provenance (lineage record)
           trialmcp-ledger (final audit)

Any State --(e-stop triggered)--> ABORTED
  Servers: safety/estop (signal propagation)
           trialmcp-ledger (abort audit record)
```

Figure 2: Robot procedure state machine with MCP server interactions per transition

2 Cross-Site MCP Communication

Source: `docs/mcp-process/02-cross-site-mcp-communication.md`

The cross-site communication model connects hospital sites, cancer centers, sponsors, and CROs through a federated coordination layer. Each site operates its own five-server deployment while participating in cross-site audit chain synchronization and token exchange protocols.

```
MULTI-SITE MCP TOPOLOGY

+------------------+     +------------------+     +------------------+
|   SITE ALPHA     |     |   SITE BETA      |     |   SITE GAMMA     |
```

```
   |  (Hospital)      |     |  (Cancer Center) |     |  (Hospital)      |
   | +------+ +-----+ |     | +------+ +-----+ |     | +------+ +-----+ |
   | |authz | |fhir | |     | |authz | |fhir | |     | |authz | |fhir | |
   | +------+ +-----+ |     | +------+ +-----+ |     | +------+ +-----+ |
   | +------+ +-----+ |     | +------+ +-----+ |     | +------+ +-----+ |
   | |dicom | |ledger||     | |dicom | |ledger||     | |dicom | |ledger||
   | +------+ +-----+ |     | +------+ +-----+ |     | +------+ +-----+ |
   | +----------+     |     | +----------+     |     | +----------+     |
   | |provenance|     |     | |provenance|     |     | |provenance|     |
   | +----------+     |     | +----------+     |     | +----------+     |
  +--------+---------+     +--------+---------+     +--------+---------+
           |                       |                       |
           +----------+-------------+-------------+----------+
                      |                           |
               +------v--------+           +------v--------+
               |   SPONSOR     |           |     CRO       |
               |  MCP Client   |           |  MCP Client   |
               +---------------+           +---------------+
```

Figure 3: Multi-site MCP topology with cross-site coordination

Cross-site token exchange enables authorized actors to access resources across sites without requiring site-specific credentials. The token exchange protocol uses scoped, time-limited tokens issued by each site's authorization server, with cross-site audit chain synchronization ensuring a unified compliance record.

3 Clinical System Integration

Source: `docs/mcp-process/03-clinical-system-integration.md`

The integration adapter architecture connects MCP servers to real hospital infrastructure through 34 adapters organized across six categories in the `integrations/` directory.

EHR Integration (FHIR R4)

- **Epic** via SMART on FHIR adapter (`integrations/fhir/smart_on_fhir.py`)

- **Cerner** via HAPI FHIR adapter (`integrations/fhir/hapi_fhir.py`)
- **MEDITECH** via terminology mapping
- De-identification pipeline: `integrations/fhir/deid_adapter.py`

Terminology Standards

- ICD-10, SNOMED CT, LOINC, RxNorm
- Mapping adapter: `integrations/fhir/terminology_adapter.py`

PACS Integration (DICOM)

- **dcm4chee**: `integrations/dicom/dcm4chee_adapter.py`
- **Orthanc**: `integrations/dicom/orthanc_adapter.py`
- **DICOMweb**: `integrations/dicom/dicomweb_adapter.py`
- Modality filter, RECIST adapter, metadata-only retrieval

Identity Providers

- OIDC adapter: `integrations/identity/oidc_adapter.py`
- mTLS adapter: `integrations/identity/mtls_adapter.py`
- KMS adapter: `integrations/identity/kms_adapter.py`

Category	Count	Key Adapters
FHIR	9	HAPI FHIR, SMART on FHIR, de-identification, terminology
DICOM	9	dcm4chee, Orthanc, DICOMweb, RECIST, modality filter
Clinical	3	Scheduling, e-consent, provenance export
Federation	4	Coordinator, policy enforcement, secure aggregation
Identity	5	OIDC, mTLS, KMS, policy engine
Privacy	4	De-identification pipeline, privacy budget, data residency

Total	**34**

4 Safety Gate Evaluation

Source: `docs/mcp-process/04-safety-gate-evaluation.md`

Five mandatory pre-procedure safety gates must pass sequentially before any robotic procedure can proceed. Each gate has a 300-second timeout. If any gate fails, the procedure transitions to ABORTED and a full audit record is created.

1. **Patient Consent** — PASS
   Verify active, informed consent for the specific procedure
2. **Site Capability** — PASS
   Validate site meets required conformance level
3. **Robot Capability** — PASS
   Confirm robot firmware and capability profile match task requirements
4. **Protocol Compliance** — PASS
   Verify procedure parameters satisfy trial protocol constraints
5. **Human Approval** — PENDING
   Require explicit clinician authorization (300s timeout)

The gate evaluation animation above demonstrates the sequential pass/fail workflow. In production, the gate service (`safety/gate_service.py`) integrates with all five MCP servers to validate each condition.

5 Federated Learning Coordination

Source: `docs/mcp-process/05-federated-learning-coordination.md`

Multi-site federated learning enables collaborative model training without centralizing patient data. The system supports three aggregation strategies with differential privacy budget tracking per site.

Aggregation Strategies

1. **FedAvg**: Weighted averaging proportional to local dataset size
2. **FedProx**: Proximal term regularization for heterogeneous data
3. **SCAFFOLD**: Variance reduction through control variates

Privacy Controls

- Differential privacy budget (epsilon) tracking per site per round
- Secure aggregation with mask generation
- Privacy adapter: `integrations/privacy/privacy_budget.py`

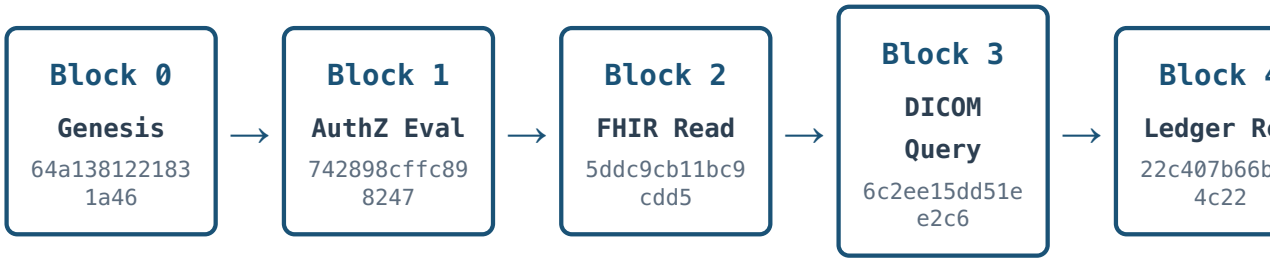
1. Coordinator distributes global model to sites
2. Each site trains locally on private data
3. Sites generate masks for secure aggregation
4. Masked model updates submitted to coordinator
5. Aggregation (masks cancel), global model updated
6. Updated model distributed, privacy budgets decremented

Federation adapters are implemented in `integrations/federation/`, including the coordinator (`coordinator.py`), policy enforcement (`policy_enforcer.py`), secure aggregation (`secure_aggregation.py`), and site harmonization (`site_harmonization.py`) modules.

6 Audit and Provenance Chain

Source: `docs/mcp-process/06-audit-provenance-chain.md`

The audit ledger maintains a hash-chained sequence of records using SHA-256 with canonical JSON serialization. Every MCP tool invocation produces an audit record linked to the prior record, creating a tamper-evident chain satisfying 21 CFR Part 11 requirements.



Demonstrative hash chain using simplified hashing. Production uses SHA-256 with canonical JSON serialization.

```
PROVENANCE DAG (Directed Acyclic Graph)

[FHIR Read: Patient/P-001]
     |
     v
[DICOM Query: CT Studies]---->[RECIST Measurement]
     |                              |
     v                              v
[Robot Procedure]------------->[Audit Record]
     |                              |
```

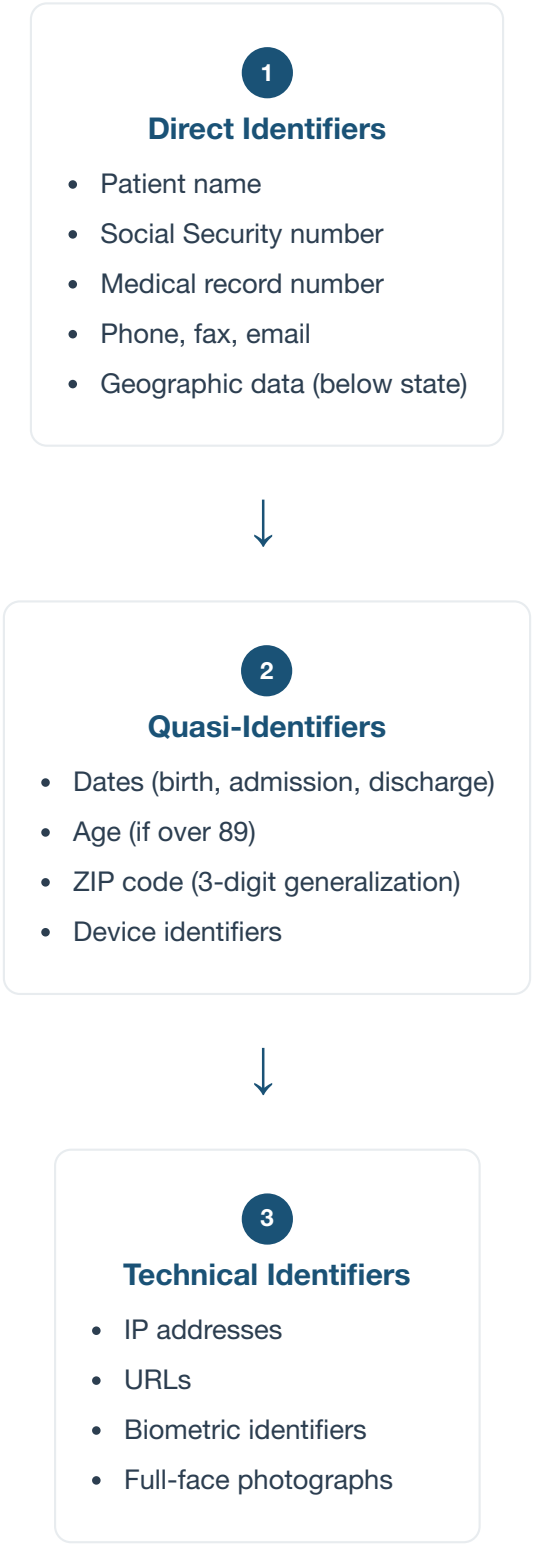
```
     v                                v
[Provenance Node]            [Chain Block #N]
     |
     v
[Forward Lineage Query]

Source types: fhir, dicom, computed, external, federated
Action types: read, transform, aggregate, transfer, derive
```

Figure 4: DAG-based provenance graph with forward/backward lineage queries

7 Privacy and De-identification

Source: `docs/mcp-process/07-privacy-deidentification.md`

The HIPAA Safe Harbor 18-identifier pipeline removes protected health information (PHI) across three stages before clinical data leaves the server boundary. HMAC-SHA256 pseudonymization preserves referential integrity across de-identified records.



Data residency enforcement ensures compliance with federal, California (CCPA/CMIA), and New York state privacy regulations through the data residency adapter (`integrations/privacy/data_residency.py`). Access control combines

RBAC and ABAC checks across four data classification levels.

National Deployment Topology

Three-tier architecture: National Governance, Regional Layers, 750+ Sites

National Governance Layer

Standards Body | Conformance Registry | Schema Registry | Validation Services

Regional Layers

Region 1 (East)

210 sites

1,050 servers

NY, MA, PA, FL, NC, VA, MD, CT, NJ, GA

Region 2 (Central)

305 sites

1,525 servers

TX, IL, OH, MI, MN, MO, IN, WI, IA, TN

Region 3 (West)

248 sites

1,240 servers

CA, WA, OR, CO, AZ, UT, NV, HI

Federated Layer

Privacy-Preserving Aggregation | Differential Privacy | Audit Chain Merge | FedAvg / FedProx / SCAFFOLD

```
NATIONAL GOVERNANCE LAYER
+-------------------------------------------------------+
| Standards Body   | Conformance Registry               |
| Extension Mgmt   | Version Compatibility Policy       |
| Schema Registry  | Validation Services                |
+--------+---------+---------+--------+--------+------+
         |                   |                 |
+--------v------+  +---------v-----+  +--------v-----+
| REGION 1      |  | REGION 2      |  | REGION N     |
```

```
| (East)        |  | (Central)     |  | (West)       |
| 200+ Sites    |  | 300+ Sites    |  | 250+ Sites   |
| 5 Svrs/Site   |  | 5 Svrs/Site   |  | 5 Svrs/Site  |
+-------+-------+  +-------+-------+  +-------+------+
        |                  |                  |
        +------------------+------------------+
                           |
                  +--------v-----------+
                  | FEDERATED LAYER    |
                  | Aggregation        |
                  | Differential Priv. |
                  | Audit Merge        |
                  +--------------------+
```

Figure 5: National 3-tier deployment topology from the paper (Section 4.6)

Deployment Simulator

Run deterministic simulations of national deployment, robot procedure workflows, and audit chain verification. All data is synthetic and modeled.

Deploy National | Robot Workflow | Verify Audit Chain | Stop

```
Click a simulation button above to begin.
```

Conformance Levels

Five hierarchical levels enabling incremental adoption (click to expand)

1

Level 1: Core (10 tools)

Basic authorization and tamper-evident audit logging

2 **Level 2: Clinical Read** (14 tools)
FHIR R4 clinical data access with mandatory HIPAA Safe Harbor de-identification

3 **Level 3: Imaging** (18 tools)
DICOM imaging queries and RECIST 1.1 tumor measurements for treatment response evaluation

4 **Level 4: Federated Site** (23 tools)
Multi-site data lineage tracking and federated learning participation

5 **Level 5: Robot Procedure** (23 tools)
Full autonomous clinical workflow with safety modules and procedure state machines

Eight conformance profiles are defined in `profiles/`, including five hierarchical profiles and three regulatory overlays (`country-us-fda.md`, `state-us-ca.md`, `state-us-ny.md`).

Safety Modules

Eight safety modules designed for Physical AI systems in clinical environments

Emergency Stop Controller

`safety/estop.py`

E-stop signal propagation, abort workflow, post-abort evidence capture, recovery with re-authorization.

IDLE → TRIGGERED → ACKNOWLEDGED → RECOVERING

Procedure State Machine

`safety/procedure_state.py`

State machine enforcement with valid transition validation across 6 states.

IDLE → PREPARING → ACTIVE → COMPLETING

Robot Registry

`safety/robot_registry.py`

Robot registration, capability validation, firmware version tracking. Capability declaration is required before any procedure.

Task Validator

`safety/task_validator.py`

Task order validation against safety constraints and robot capabilities using `schemas/task-order.schema.json`.

Safety Gate Service

`safety/gate_service.py`

5 mandatory gates with sequential evaluation: Patient Consent, Site Capability, Robot Capability, Protocol Compliance, Human Approval. 300-second timeout per gate.

Approval Checkpoint

`safety/approval_checkpoint.py`

Multi-party approval requiring concurrent authorization from clinician, system, and regulatory parties before critical procedure transitions.

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Site Verifier

`safety/site_verifier.py`

Site capability verification against conformance requirements and infrastructure readiness checks.

Boundary Enforcer

`safety/boundary_enforcer.py`

Execution boundary enforcement: spatial boundary validation, force limit enforcement, temporal constraint checking.

Emergency Stop Code (from `safety/estop.py`)

```
class EStopStatus(Enum):
    IDLE = "IDLE"
    TRIGGERED = "TRIGGERED"
    ACKNOWLEDGED = "ACKNOWLEDGED"
    RECOVERING = "RECOVERING"

class EStopController:
    def trigger_estop(self, procedure_id, reason,
                      triggered_by, current_state=None):
        if self._status == EStopStatus.TRIGGERED:
            raise RuntimeError("E-stop already active")
        signal = EStopSignal(
```

```
            signal_id=str(uuid.uuid4()),
            procedure_id=procedure_id,
            reason=reason,
            triggered_by=triggered_by,
            affected_servers=list(self._active_servers),
            preserved_state=current_state or {})
        self._active_signal = signal
        self._status = EStopStatus.TRIGGERED
        return signal
```

Repository Scale

Quantitative summary of the national repository at v1.2.0

Category	Count
MCP servers	5
Tools (total)	23
JSON schemas	13
Conformance profiles	8
Test functions	668
Integration adapters	34
Specification docs	9
ADRs	7
Total files	381+

Lines of code (approx.)	69,800+

Category	Count
Safety modules	8
Benchmark modules	5
Certification tools	4
Deployment modes	6
Interop scenarios	8
Actor personas	6
Stakeholder guides	5
Operational docs	5
Total directories	88
Regulatory overlays	7

Test Coverage Breakdown

Category	Tests	Validates
Unit (integrations)	205	All 34 integration adapters
Unit (safety)	88	All 8 safety modules
Unit (core server)	32	5 MCP server implementations
Unit (infrastructure)	12	Conformance harness, schema validator

Positive conformance	69	Correct behavior at Levels 1-3
Adversarial	57	AuthZ bypass, chain tampering, PHI leakage
Blackbox	63	Per-server external conformance
Security	39	SSRF prevention, token lifecycle, chain integrity
Interoperability	32	Cross-server trace, schema validation
Negative	26	Invalid input rejection, unauthorized access
Integration	13	Multi-server integration scenarios
Total	**668**	

CI/CD Pipeline (14 Jobs)

The repository maintains quality through a GitHub Actions CI pipeline defined in `.github/workflows/ci.yml`:

1. `lint-and-format`: Ruff linting and formatting (Python 3.10, 3.11, 3.12)
2. `integration-tests`: Multi-server integration validation
3. `adversarial-tests`: Security attack resistance
4. `schema-validation`: JSON Schema Draft 2020-12 validation
5. `schema-compatibility`: Schema metadata and field consistency
6. `contract-consistency`: Generated model alignment with schemas
7. `benchmark-smoke`: Performance benchmark verification
8. `sdk-python`: Python SDK import verification
9. `sdk-typescript`: TypeScript SDK compilation
10. `cli-smoke`: CLI command validation
11. `codegen-consistency`: Code generation pipeline verification
12. `security-scan`: Dependency audit and secret detection
13. `typescript-build`: Reference TypeScript compilation

14. `docs-lint` : Documentation completeness and link integrity

Adoption Roadmap

Stakeholder-specific pathways for implementing the national standard

S

Sponsors and CROs

Begin with Level 1 (Core) deployment for centralized audit trail access. Progress to Level 2 for de-identified clinical data queries. Guide: `docs/guides/sponsor-cro.md`

H

Hospital IT

Deploy five-server stack using Docker Compose (`deploy/docker-compose.yml`). Integrate with existing EHR via FHIR adapters and PACS via DICOM adapters. Guide: `docs/guides/hospital-it.md`

R

Regulators and IRBs

Review conformance profiles and regulatory overlays (`profiles/`). Evaluate audit chain integrity using `ledger_verify` . Guide: `docs/guides/regulator-irb.md`

V

Robot Vendors

Implement MCP client interface against the 23-tool contract. Register robots via `safety/robot_registry.py` . Run conformance tests at Level 5. Guide: `docs/guides/robot-vendor.md`

C

Standards Community

Review the 9 normative specifications in `spec/`, 7 ADRs in `docs/adr/`, and governance framework. Guide: `docs/guides/standards-community.md`

Robot Procedure Data Flow

```
ROBOT AGENT         MCP SERVERS         CLINICAL SYSTEMS
-----------         -----------         ----------------
+--------+
| START  |
+---+----+
    |  1. Request Token
    v
+--------+        +--------------+
| AuthZ  |--------| trialmcp-    |       Issue scoped token
| Phase  |--------| authz        |       (role + expiry)
+---+----+        +--------------+
    |  2. Query Patient Data
    v
+--------+        +--------------+       +-----------+
| FHIR   |--------| trialmcp-    |-------| EHR       |
| Phase  |--------| fhir         |-------| (FHIR R4) |
+---+----+        +--------------+       +-----------+
    |  3. Retrieve Imaging
    v
+--------+        +--------------+       +-----------+
| DICOM  |--------| trialmcp-    |-------| PACS      |
| Phase  |--------| dicom        |-------| (DICOM)   |
+---+----+        +--------------+       +-----------+
    |  4. Execute Procedure
    v
+--------+
| ROBOT  |                               Physical AI task
| ACTION |                               under safety rules
+---+----+
    |  5. Record Audit
    v
+--------+        +--------------+
| Audit  |--------| trialmcp-    |       Hash-chained record
| Phase  |--------| ledger       |       (21 CFR Part 11)
+---+----+        +--------------+
    |  6. Record Provenance
```

```
    v
+--------+        +--------------+
| Prov.  |--------| trialmcp-    |       DAG lineage entry
| Phase  |--------| provenance   |       (SHA-256)
+---+----+        +--------------+
    v
+--------+
|  DONE  |
+--------+
```

Figure 6: Complete data flow for a Physical AI robot agent executing a clinical procedure

Classification and Status

What This IS

A **reference architecture demonstration and interactive simulation** of the proposed National MCP-PAI Oncology Trials Standard. It serves as an educational and stakeholder communication tool that visualizes the system described in the paper using deterministic simulations with pre-configured data.

What This Is NOT

- NOT a deployed clinical system
- NOT approved for use in actual oncology trials
- NOT connected to real patient data, EHR systems, PACS, or robotic systems
- NOT a medical device or software as a medical device (SaMD)

Simulation Conditions and Approximations

- All data shown is synthetic
- Network latencies, site counts, and response times are modeled approximations based on the paper's architecture
- Federated learning rounds use illustrative aggregation, not real model weights
- Hash chains demonstrate the algorithm with sample data

What Is Needed for Actual Trial Use

1. FDA review and recognition as a consensus standard
2. Clinical pilot deployment at NCI-designated cancer centers
3. Real EHR/PACS adapter validation with vendor systems (Epic, Cerner, Orthanc, dcm4chee)
4. Robot vendor certification program
5. IRB approval at each participating site
6. 21 CFR Part 11 compliance audit of deployed instances
7. HIPAA security risk assessment

Current Status

This is a **proposed reference standard** with a complete reference implementation (381 files, approximately 69,800 lines) that has not yet been deployed in clinical settings. The repository (v1.2.0) includes the complete specification, reference implementations in Python and TypeScript, a 668-function test suite, and a 20-page peer-reviewed paper.

Citation and Links

Cite This Work

Kawchak K. National MCP Servers for Physical AI Oncology Clinical Trial Systems. Zenodo. 2026; 10.5281/zenodo.18916731.

BibTeX

```
@article{kawchak2026national,
  title     = {National MCP Servers for Physical AI
               Oncology Clinical Trial Systems},
  author    = {Kawchak, Kevin},
  year      = {2026},
  publisher = {Zenodo},
  doi       = {10.5281/zenodo.18916731},
  url       = {https://doi.org/10.5281/zenodo.18916731}
}
```

Repository Links

Repository	DOI	Focus
national-mcp-pai-oncology-trials	10.5281/zenodo.18894758	National standard, SDKs, governance (current)
mcp-pai-oncology-trials	10.5281/zenodo.18869776	MCP server proof of concept, 5-server architecture
pai-oncology-trial-fl	10.5281/zenodo.18840880	Federated learning, privacy modules
physical-ai-oncology-		USL scoring, single-site Physical AI

trials	10.5281/zenodo.18445179	framework

Paper

- **PDF**: National MCP Servers for Physical AI Oncology Clinical Trial Systems
- **LaTeX Source**: Latex Source Files.zip
- **Template**: kourgeorge/arxiv-style

Contributors

This demonstration was generated by **Anthropic Claude Code Opus 4.6** and **OpenAI ChatGPT 5.4 Thinking** under the direction of Kevin Kawchak.

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